

## Project Planning

|              |                                                             |
|--------------|-------------------------------------------------------------|
| Date         | 20 July 2026                                                |
| Project Name | OptiCrop: Smart Agricultural Production Optimization Engine |

### Product Backlog, Sprint Schedule, and Estimation

| Sprint   | Functional Requirement (Epic)            | User Story Number | User Story / Task                                                                              | Story Points | Priority | Team Members           | Sprint Start Date | Sprint End Date (Planned) |
|----------|------------------------------------------|-------------------|------------------------------------------------------------------------------------------------|--------------|----------|------------------------|-------------------|---------------------------|
| Sprint-1 | Epic 1: Define Problem and Understanding | USN-1             | As a team, we specify the business problem 2 farmers face in crop selection                    | 2            | High     | Jagadeeswari, Manjusha | [date]            | [date]                    |
| Sprint-1 | Epic 1                                   | USN-2             | As a team, we document business requirements (functional & non-functional)                     | 2            | High     | Jagadeeswari           | [date]            | [date]                    |
| Sprint-1 | Epic 1                                   | USN-3             | As a team, we conduct a literature survey on existing crop recommendation systems              | 1            | Medium   | Manjusha               | [date]            | [date]                    |
| Sprint-1 | Epic 1                                   | USN-4             | As a team, we assess the social and business impact of the solution                            | 1            | Medium   | Jagadeeswari           | [date]            | [date]                    |
| Sprint-2 | Epic 2: Data Collection and Analysis     | USN-5             | As a developer, I download the crop recommendation dataset                                     | 1            | High     | Jagadeeswari           | [date]            | [date]                    |
| Sprint-2 | Epic 2                                   | USN-6             | As a developer, I import required libraries (pandas, numpy, matplotlib, seaborn, scikit-learn) | 1            | High     | Jagadeeswari           | [date]            | [date]                    |
| Sprint-2 | Epic 2                                   | USN-7             | As a developer, I read and load the dataset into a DataFrame                                   | 1            | High     | Jagadeeswari           | [date]            | [date]                    |
| Sprint-2 | Epic 2                                   | USN-8             | As a developer, I perform univariate analysis on each feature                                  | 2            | Medium   | Manjusha               | [date]            | [date]                    |
| Sprint-2 | Epic 2                                   | USN-9             | As a developer, I perform bivariate analysis between feature pairs                             | 2            | Medium   | Manjusha               | [date]            | [date]                    |
| Sprint-2 | Epic 2                                   | USN-10            | As a developer, I perform multivariate analysis across                                         | 2            | Medium   | Jagadeeswari           | [date]            | [date]                    |

|          |                              |        |                                                                                      |   |        |                        |        |        |
|----------|------------------------------|--------|--------------------------------------------------------------------------------------|---|--------|------------------------|--------|--------|
| Sprint-3 | Epic 3: Data Pre-processing  | USN-11 | all features<br>As a developer, I handle missing/duplicate values in the dataset     | 1 | High   | Jagadeeswari           | [date] | [date] |
| Sprint-3 | Epic 3                       | USN-12 | As a developer, I encode the categorical crop label                                  | 1 | High   | Manjusha               | [date] | [date] |
| Sprint-3 | Epic 3                       | USN-13 | As a developer, I scale/normalize numerical features                                 | 1 | Medium | Jagadeeswari           | [date] | [date] |
| Sprint-3 | Epic 3                       | USN-14 | As a developer, I split the dataset into train and test sets                         | 1 | High   | Manjusha               | [date] | [date] |
| Sprint-4 | Epic 4: Model Building       | USN-15 | As a developer, I train a K-Means Clustering model to explore soil/climate groupings | 2 | Medium | Jagadeeswari           | [date] | [date] |
| Sprint-4 | Epic 4                       | USN-16 | As a developer, I train a Logistic Regression model for crop classification          | 2 | High   | Manjusha               | [date] | [date] |
| Sprint-4 | Epic 4                       | USN-17 | As a developer, I evaluate model performance and save the best model                 | 2 | High   | Jagadeeswari           | [date] | [date] |
| Sprint-4 | Epic 4                       | USN-18 | As a user, I get the best crop predicted based on given parameters                   | 2 | High   | Manjusha               | [date] | [date] |
| Sprint-5 | Epic 5: Application Building | USN-19 | As a developer, I build the HTML pages for the input form and results                | 2 | High   | Jagadeeswari           | [date] | [date] |
| Sprint-5 | Epic 5                       | USN-20 | As a developer, I build the Python (Flask) backend code                              | 3 | High   | Manjusha               | [date] | [date] |
| Sprint-5 | Epic 5                       | USN-21 | As a developer, I run and test the complete application end-to-end                   | 2 | High   | Jagadeeswari, Manjusha | [date] | [date] |

*Note: Epics 6–9 (e.g., Testing, Documentation, Demonstration, and Conclusion) were tracked as final-phase activities and are detailed in folders [6. Project Testing](#), [7. Project Documentation](#), and [8. Project Demonstration](#) of this repository — add their specific story rows here (USN-22 to USN-25) once finalized in SkillWallet.*

### Sprint Velocity Summary

| Sprint   | Total Story Points | Stories Planned | Stories Completed |
|----------|--------------------|-----------------|-------------------|
| Sprint-1 | 6                  | 4               | [ ]               |
| Sprint-2 | 9                  | 6               | [ ]               |
| Sprint-3 | 4                  | 4               | [ ]               |
| Sprint-4 | 8                  | 4               | [ ]               |
| Sprint-5 | 7                  | 3               | [ ]               |

